

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION)

CLASS: BTECH
BRANCH: CSE/AI ML/ECE/EEE

SEMESTER: I
SESSION : MO/2023

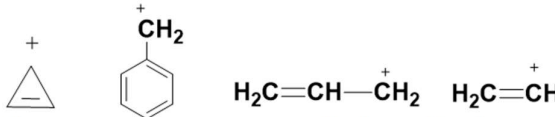
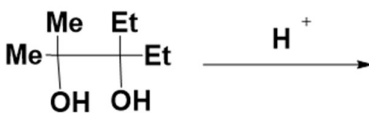
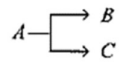
SUBJECT: CH101R1 CHEMISTRY

TIME: 3 Hours

FULL MARKS: 50

INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Before attempting the question paper, be sure that you have got the correct question paper.
5. Tables/Data hand book/Graph paper etc. to be supplied to the candidates in the examination hall.

- | | [3+2] | CO | BL |
|--|-------|----|----|
| Q.1(a) For the complex ion $[\text{Fe}(\text{Cl})_6]^{3-}$, determine the number of d electrons for Fe, sketch the crystal field splitting of d-orbital energy levels. List the number of unpaired electrons, and label whether the complex is paramagnetic or diamagnetic? | [3+2] | 1 | 2 |
| Q.1(b) What are 'spin forbidden' and 'spin allowed' transitions? Why is $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$ colourless while MnO_4^- intensely colored? | [3+2] | 1 | 2 |
| Q.2(a) (i) Write the correct order of stability of the following carbocations with an explanation: | [2+3] | 2 | 2 |
|  | | | |
| (ii) Draw the Fischer, Sawhorse, Newman, Projections formula of $(\text{CH}_3)\text{CHOH}-(\text{CH}_3)\text{CHOH}$ and identify the most stable conformations. | | | |
| Q.2(b) (i) Arrange increasing order of SN^1 and SN^2 reaction of the following compounds with an explanation: CH_3Br , $\text{CH}_3\text{CH}_2\text{Br}$, $(\text{CH}_3)_2\text{CHBr}$, $(\text{CH}_3)_3\text{Br}$ | [3+2] | 2 | 3 |
| (ii) Predict the products with plausible mechanism, in the following case: | | | |
|  | | | |
| Q.3(a) (i) Draw and explain the concentrations vs time plot for reactant and products of parallel reactions. | [3+2] | 3 | 2 |
|  | | | |
| (ii) The half-lives of two parallel path reaction are 4 hrs and 12 hrs respectively, calculate the average half-life for the decay of A. | | | |
| Q.3(b) (i) Draw 'rate' vs $[\text{S}]$ plot of Michaelis-Menten equation and explain the different parts of it. How Michaelis-Menten constant (K_m) and maximum rate (V_{max}) can be determined? | [3+2] | 3 | 2 |
| (ii) Explain the working principle of phase-transfer catalyst along with the example. | | | |
| Q.4(a) State the Beer-Lambert law. The transmittance of an aqueous solution of KMnO_4 at a certain wavelength is 1% for 10^{-4} molar solution in a 1 cm cell. What is its absorbance and the molar absorption coefficient of KMnO_4 ? | [2+3] | 4 | 2 |
| Q.4(b) (i) How many vibrational modes are there in C_6H_6 , H_2O and CO_2 ? | [3+2] | 4 | 3 |
| (ii) Draw the NMR spectra by indicating the number of peaks for each proton in $\text{CH}_3\text{CH}_2\text{OH}$ and $\text{CH}_3\text{CH}_2\text{CH}_2\text{Br}$. | | | |
| Q.5(a) Calculate the number of components for decomposition of NH_4Cl (s) in presence of excess NH_3 . Draw the phase diagram of water with proper explanation. | [2+3] | 5 | 2 |
| Q.5(b) Write down the anodic, cathodic and overall reactions for (i) Li-ion batteries and (ii) H_2 - O_2 fuel cell operated in acidic medium. | [3+2] | 5 | 2 |